Homework 2

Machine Learning & AI Part I

Theory, experiments, and results

1. Predicting trip duration and observation uncertainty

(a) Link function

Write the two real-valued outputs of the architecture as $g(x) = (a(x), b(x))$. An appropriate link is

$$f_\mu(x) = a(x), \qquad f_\sigma(x) = \text{softplus}(b(x)) + \varepsilon = \log(1 + e^{b(x)}) + \varepsilon, \qquad \varepsilon > 0.$$

The identity link leaves the mean unrestricted, while softplus ensures a positive standard deviation. The small floor prevents division by zero and unstable logarithms. An exponential link for the standard deviation would also be valid. The uncertainty modeled here is observation noise conditional on X, rather than uncertainty about the fitted network parameters.

(b) Gaussian negative log-likelihood

Under $Y \mid X = x \sim \mathcal{N}(f_\mu(x), f_\sigma(x)^2)$, the negative log-likelihood per observation is

$$-\log p(y_i \mid x_i) = \frac{1}{2}\log(2\pi) + \log f_\sigma(x_i) + \frac{(y_i - f_\mu(x_i))^2}{2 f_\sigma(x_i)^2}.$$

Thus minimize the average loss

$$\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left[\log f_\sigma(x_i) + \frac{(y_i - f_\mu(x_i))^2}{2 f_\sigma(x_i)^2} + \frac{1}{2}\log(2\pi)\right].$$

Increasing the predicted variance reduces the residual penalty but increases the $\log f_\sigma$ term, so the network cannot improve the objective simply by making every standard deviation arbitrarily large.

(c) Model, splits, and regression evaluation

The taxi dataset contains 523,554 rows, seven supplied standardized features, and trip duration in minutes. A reproducible random row split, with seed 42, produces 366,487 training, 78,533 validation, and 78,534 test observations (approximately 70/15/15). No rows are removed.

The network is $7 \to 64 \to 64 \to 2$, with ReLU after each hidden layer and the links in part (a). For numerical stability, duration is centered and scaled using only the training mean $m = 10.4163$ and standard deviation $s = 7.7553$. For a standardized-target output $(\widetilde{\mu}, \widetilde{\sigma})$, predictions in minutes are $\mu = m + s\widetilde{\mu}$ and $\sigma = s\widetilde{\sigma}$. The positive floor is 10^{-3} in standardized units. Training on this target changes the original-unit Gaussian NLL only by the constant $\log s$.

AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, batch size 2,048, and gradient-norm clipping at 10. Validation NLL selects the checkpoint, with patience 8 and a 70-epoch cap. Training stops after 54 epochs and restores epoch 46. The test split is used for the final evaluation only. A random row split assesses similar-distribution trips, rather than a future-time forecasting task.

1. Trip duration: evaluation and probabilities (continued)

Two regression baselines are fitted on training data: a constant Gaussian using the training mean and standard deviation, and linear regression with a constant standard deviation estimated from its training residuals. All NLL values below are in original minute units, including the Gaussian constant.

Model	Test Gaussian NLL	Test RMSE (minutes)
Constant Gaussian	3.46752	7.75689
Linear mean, constant variance	3.43975	7.54446
Gaussian MLP	**2.74665**	**4.00621**

Table 1: The conditional mean/variance MLP improves on both regression baselines.

(d) Probability that a trip takes less than 45 minutes

Let Φ denote the standard normal CDF. Standardizing the conditional normal variable gives

$$\boxed{\Pr(Y < 45 \mid X = x) = \Phi\left(\frac{45 - f_\mu(x)}{f_\sigma(x)}\right).}$$

The mean, standard deviation, and threshold must all use the same units.

(e) Probability evaluation

For $z_i = \mathbf{1}\{y_i < 45\}$ and the probabilities p_i above, evaluate

$$\text{Brier} = \frac{1}{n_{\text{test}}} \sum_i (p_i - z_i)^2, \qquad \text{Accuracy} = \frac{1}{n_{\text{test}}} \sum_i \mathbf{1}\{\mathbf{1}[p_i \geq 0.5] = z_i\}.$$

Also report Bernoulli log-loss, $-n_{\text{test}}^{-1} \sum_i [z_i \log p_i + (1 - z_i) \log(1 - p_i)]$. For its numerical evaluation only, probabilities are clipped to $[10^{-12}, 1 - 10^{-12}]$ in double precision.

Model	Brier score ↓	Log-loss ↓	Accuracy ↑
Constant Gaussian	0.004457	0.055276	99.5543%
Linear Gaussian	0.004455	0.045934	99.5543%
Training event-frequency baseline	0.004437	0.028572	99.5543%
Gaussian MLP	**0.002861**	**0.022725**	**99.6702%**

Table 2: Probability scores on 78,534 held-out taxi trips.

Only 350 test trips take at least 45 minutes; 99.5543% take less than 45. Consequently, an always-under-45 classifier already has very high accuracy. The MLP's improvement in Brier score and log-loss over the empirical event-rate baseline is stronger evidence of useful probabilities than accuracy alone. Its nominal 90% Gaussian prediction intervals cover 93.9631% of test durations, suggesting conservative aggregate coverage. Gaussianity is still an approximation: it permits negative durations and need not capture tail shape or conditional calibration accurately.

1. Training and uncertainty diagnostics

The learning curves show improvement in both training and validation Gaussian NLL, with checkpoint selection based on validation loss rather than the final epoch. The plotted NLL uses standardized targets; the reported original-minute NLL adds the constant $\log s$.

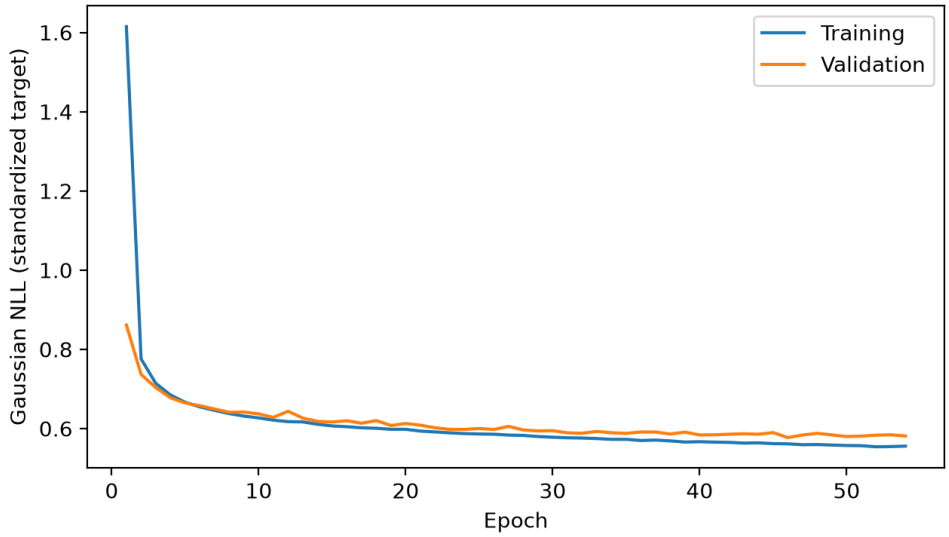


Figure 1: Taxi training and validation loss across epochs. The selected checkpoint is epoch 46; training ends at epoch 54 after eight epochs without sufficient validation improvement.

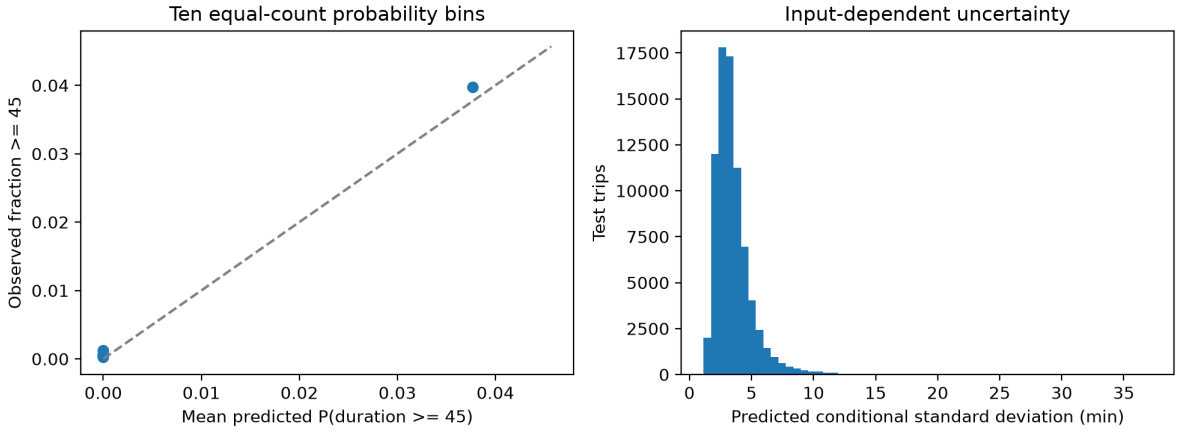


Figure 2: Left: calibration of the rare complementary event, duration ≥ 45, in ten equal-count probability bins. The diagonal indicates perfect calibration. Right: variation in predicted conditional standard deviation.

2. Encoding categorical retail features

(a) Cardinality, feasibility, and data splitting

There are 397,924 retail transactions. Treating the identifiers as categorical strings gives 3,665 distinct `StockCode` values and 4,339 distinct `CustomerID` values. One-hot encoding would therefore create 8,004 columns before any reference-category removal. A dense double-precision matrix would occupy about 25.5 GB, but sparse one-hot encoding is feasible: each row has just two nonzero entries. Learned embeddings offer a compact alternative and can share information between categories through a low-dimensional representation.

A seed-42 random row split creates 278,546 training, 59,689 validation, and 59,689 test observations. All three models use these same splits. The training split contains 3,595 stock codes and 4,311 customer IDs. In the test split, 0.0854% of stock codes and 0.0268% of customer IDs are unseen in training. No validation or test targets are used to build training encodings.

Quantity has median 6, 90th percentile 24, 99th percentile 120, 99.9th percentile 504, and maximum 80,995. To reduce the influence of extreme transactions, fit models to $t_i = \log(1 + y_i)$ using squared error. If the model predicts $h(x)$, use $\widehat{y}(x) = \exp(\max\{h(x), 0\}) - 1$. The main evaluation metric is

$$\text{MSLE} = \frac{1}{n}\sum_{i=1}^{n}\big[\log(1+y_i) - \log(1+\widehat{y}_i)\big]^2.$$

MAE and RMSE on the original quantity scale are secondary metrics. This objective targets typical multiplicative behavior: exponentiating a conditional mean of log-quantity does *not* generally recover $\mathbb{E}[Y \mid X]$. Continuous predictions are retained rather than rounded to counts.

(b) Target encoding

For a category c, the basic target encoding is the training mean $\sum_{i:x_i=c} y_i/n_c$. Use a smoothed version to stabilize rare categories:

$$\text{TE}(c) = \frac{\sum_{i:x_i=c} y_i + \alpha\bar{y}_{\text{train}}}{n_c + \alpha}, \qquad \alpha = 20.$$

An unseen category receives the relevant training mean. The means above use *raw Quantity*, as requested, even though the regression target is $\log(1 + \text{Quantity})$.

For training features, five-fold cross-fitting computes each row's encoding using the other four folds, excluding its own target. Validation and test features use category maps fitted to the entire training split. Next apply $\log(1+\text{TE})$ to compress extreme encoded means and standardize using training-feature statistics only. This produces two numeric predictors.

Fit a histogram gradient boosting regressor to the log-target with squared error, 200 iterations, learning rate 0.05, at most 15 leaves per tree, and L_2 regularization 10; internal early stopping is disabled. These settings are fixed before test evaluation. The model obtains validation MSLE 0.400522 and test MSLE **0.401474**, compared with test MSLE 0.996641 for a constant training-mean log-target predictor. Its test MAE is 7.0012 and RMSE is 33.1750.

2. Retail embeddings and model comparison (continued)

(c) Neural embeddings

Each feature has a separate learned 16-dimensional embedding table. Concatenate the stock and customer vectors to obtain 32 inputs, then use a $32 \to 64 \to 32 \to 1$ MLP with ReLU hidden activations and a linear log-target output. Known categories use training-fitted integer IDs; unseen categories map to a fixed zero vector. There are $16(3595 + 4311) = 126{,}496$ learned category-vector parameters, excluding the fixed unknown rows.

Both retail neural models use AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch size 2,048, gradient clipping at 10, a 50-epoch cap, and validation-loss early stopping with patience 6. The embedding-only network selects epoch 48 of 50. Test MSLE is **0.293938**, substantially below target encoding alone.

(d) Combining target encoding and embeddings

Concatenate the same two standardized target-encoded features with the two embeddings. This changes only the first input width, from 32 to 34. The combined network selects epoch 41 and stops at epoch 47. Its test MSLE is **0.295851**: better than target encoding alone, but slightly worse than embeddings alone under the primary metric.

Model	Val. MSLE	Test MSLE	Test MAE	Test RMSE
Log-mean baseline	1.010325	0.996641	10.2061	41.3628
Target encoding + boosting	0.400522	0.401474	7.0012	33.1750
Embeddings + MLP	**0.291406**	**0.293938**	**6.1957**	29.9076
Combined MLP	0.292016	0.295851	6.2658	**29.7773**

Table 3: Shared held-out splits; lower values are better for every metric.

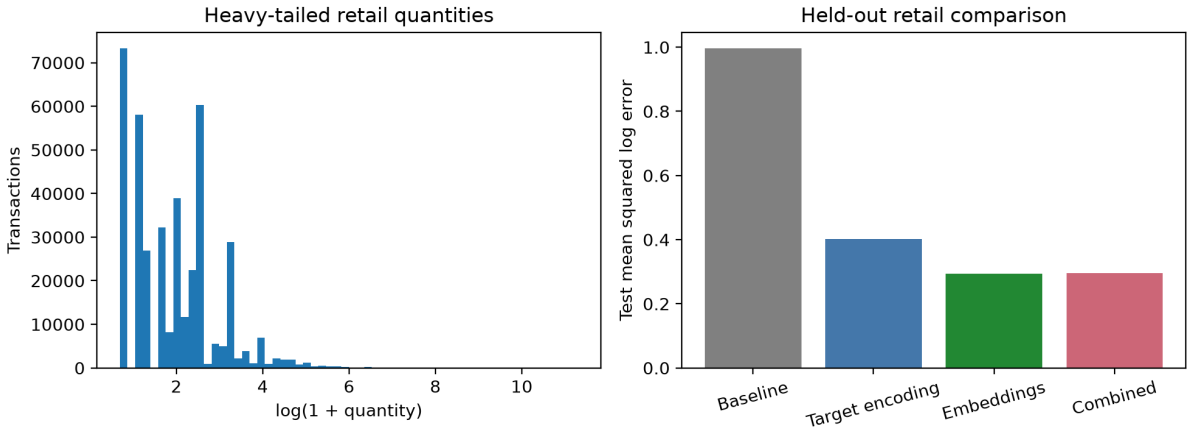


Figure 3: Quantity distribution after log transformation and the four test MSLEs.

2. What the learned representations capture

(e) Interpretation and limitations

An embedding is a learned category-specific vector. Training adjusts these vectors so that the downstream network can exploit similarities in customer purchase patterns and product behavior. Unlike a single target mean, a vector can encode several aspects of a category; the nonlinear network can use customer–product interactions. The lower held-out MSLE supports the value of these representations in this dataset.

Target encoding supplies a strong scalar summary, but it can discard useful variation and is sensitive to rare categories and extreme quantities. Adding it to embeddings can be redundant; here it does not improve MSLE. The combined model has slightly lower RMSE, illustrating that the ranking depends on the loss. These small embedding/combined differences are descriptive results for one split and seed, without a statistical significance claim. Random row splitting evaluates transactions largely involving known customers and products; it does not establish cold-start or future-time performance. The training split contains the largest quantity (80,995), while the test maximum is 3,114. Thus the raw-scale errors do not establish accuracy for transactions as extreme as the largest training example.

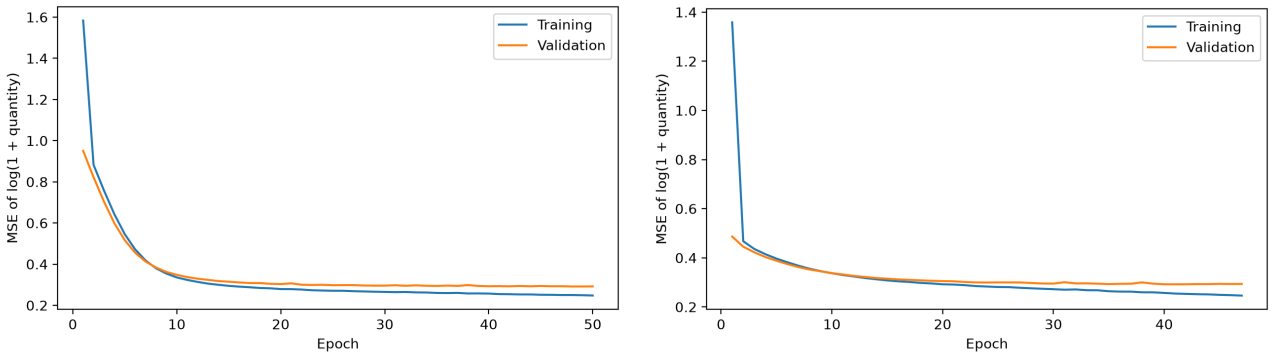


Figure 4: Retail neural-network learning curves. Validation loss selects epoch 48 for embeddings and epoch 41 for the combined model. Both curves use squared error on the log-transformed target.

The embedding-only model reaches the 50-epoch budget; the combined model stops after six epochs without a sufficient validation improvement. Both restore the best validation checkpoint. A larger training budget or different regularization could change the comparison; no additional settings are chosen using the test results.

3. Random Kitchen Sinks and a one-hidden-layer MLP

(a) Visualizing the supplied training data

The supplied splits contain 600 training, 200 validation, and 200 unlabeled test observations. Both covariates lie in $[-3, 3]$. The colored scatter below shows a smooth response pattern plus observation noise: values are low near $x_1 \approx -1.5$, $x_2 \approx 0$, rise for large $|x_2|$, and vary smoothly with x_1.

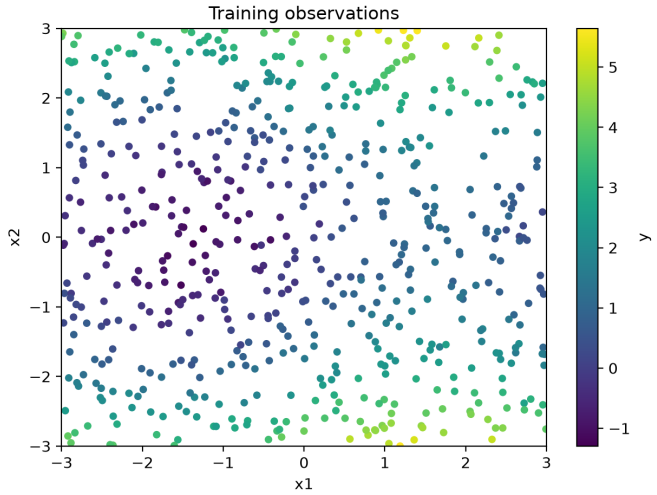


Figure 5: Training observations colored by the observed response y.

(b) RKS with ridge regression

For $h = D/2$, draw $w_j \sim \mathcal{N}(0, I/\sigma^2)$ and $b_j \sim U[0, 2\pi]$, and form the unnormalized features

$$z(x) = \big(\sin(w_1^T x + b_1), \cos(w_1^T x + b_1), \ldots, \sin(w_h^T x + b_h), \cos(w_h^T x + b_h)\big)^T.$$

Append an intercept to obtain $\phi(x)$. The ridge fit solves

$$\widehat{\beta} = \arg\min_{\beta} \|y - \Phi\beta\|_2^2 + \lambda \sum_{j=1}^{D} \beta_j^2, \quad (\Phi^T\Phi + \lambda P)\widehat{\beta} = \Phi^T y, \quad P = \mathrm{diag}(1, \ldots, 1, 0).$$

The intercept is unpenalized. The implementation uses a linear solve rather than an explicit inverse; at $\lambda = 0$ it uses least squares. Fix $\sigma = 1$, $\lambda = 10^{-3}$, and seed 42; sweep $D \in \{16, 32, 64, 128, 256, 512, 1024\}$. Select $D = 32$ ($h = 16$) using the smallest validation MSE. All other RKS settings remain fixed.

3. RKS feature-count sweep

D	Training MSE	Validation MSE
16	0.136344	0.136949
32	0.033244	**0.035983**
64	0.031591	0.037268
128	0.030544	0.038259
256	0.029458	0.041707
512	0.029127	0.044339
1024	0.028745	0.044198

Table 4: RKS feature-count sweep.

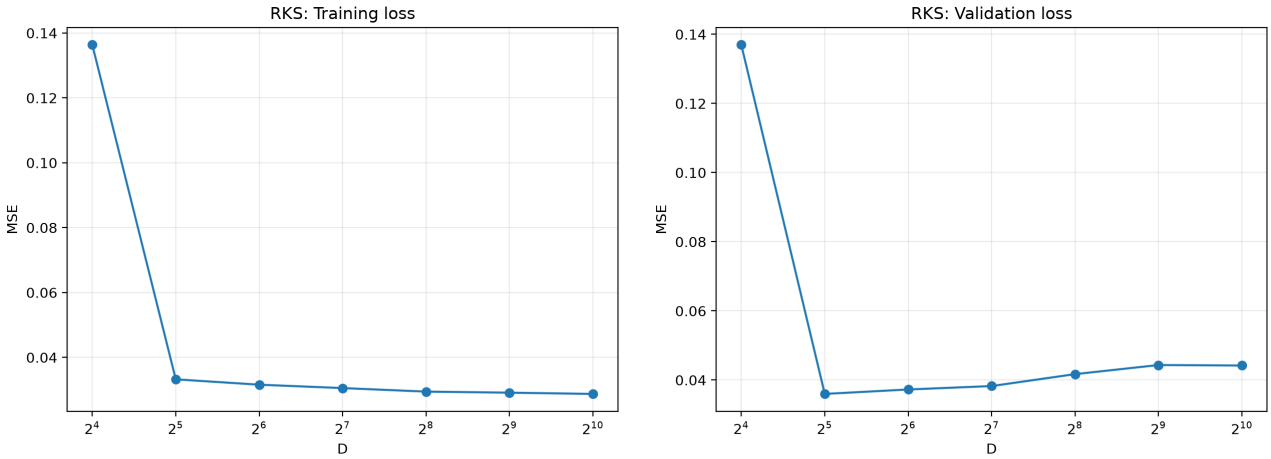


Figure 6: Separate training and validation losses for RKS. Increasing the number of features past 32 lowers training loss but increases validation loss under the fixed ridge penalty, consistent with overfitting.

3. One-hidden-layer MLP sweep

(c) One-hidden-layer MLP

Use the network

$$\widehat{f}(x) = v^T \tanh(Wx + b) + c, \qquad W \in \mathbb{R}^{m\times 2}, \quad v, b \in \mathbb{R}^m.$$

Its only hidden layer uses tanh; the output is linear. Minibatch Adam minimizes MSE for 1,500 epochs at learning rate 10^{-3} and batch size 128. Keep these settings and seed 42 fixed, varying only hidden width $m \in \{16, 32, 64, 128, 256\}$. All validation losses are measured after training; no validation rows enter gradient updates. The selected width is 256.

Hidden width m	Training MSE	Validation MSE
16	0.038742	0.041707
32	0.035296	0.037840
64	0.033872	0.037653
128	0.033128	0.037418
256	0.032513	**0.037257**

Table 5: MLP width sweep at a fixed training budget.

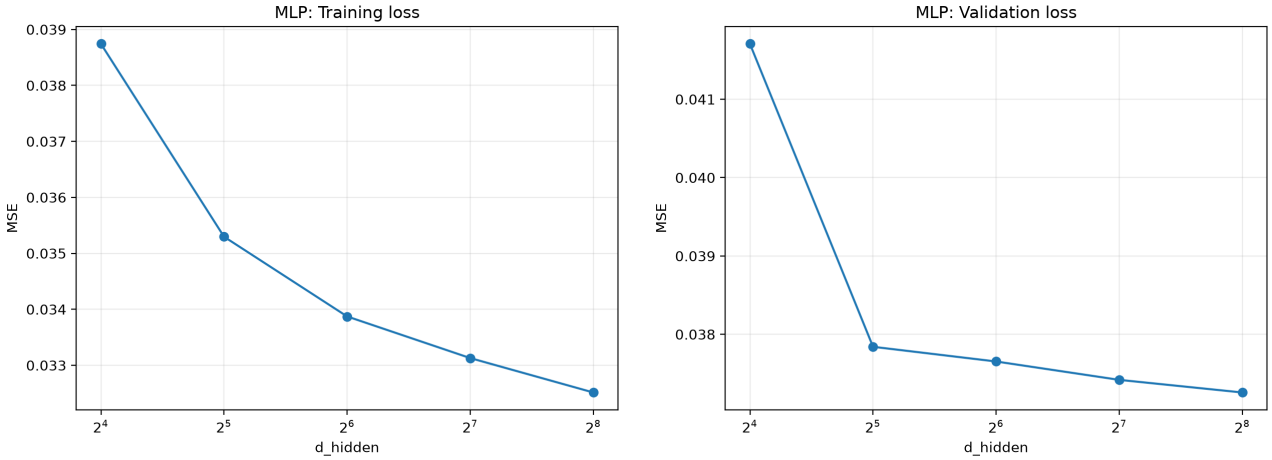


Figure 7: Separate training and validation losses for the MLP. Both improve across the tested widths; this does not establish that still larger networks would necessarily improve validation performance.

3. Comparing the selected fitted functions

(d) Final refits, predictions, and interpretation

Refit each selected configuration on the supplied training set, retaining the same seed and training settings. Keep validation rows separate so that the reported validation losses are not training-set losses. Because these losses were used for hyperparameter selection, they are not independent test error estimates.

Model	Selected setting	Training MSE	Validation MSE
RKS	$D = 32$, $\sigma = 1$, $\lambda = 10^{-3}$	0.033244	**0.035983**
MLP	Width 256, tanh	0.032513	0.037257

Table 6: Final refits reproduce the selected sweep losses. The training-mean constant baseline has validation MSE 2.396572.

RKS performs slightly better on validation in this run, although the MLP has lower training loss. The smooth sinusoidal random features provide a useful basis for this response, and ridge regression controls their fitted weights. The MLP learns its feature map jointly with the output coefficients. Its additional flexibility does not yield a validation improvement at the tested widths and training budget. The experiment supports this comparison for these settings, rather than a universal ranking of the two approaches.

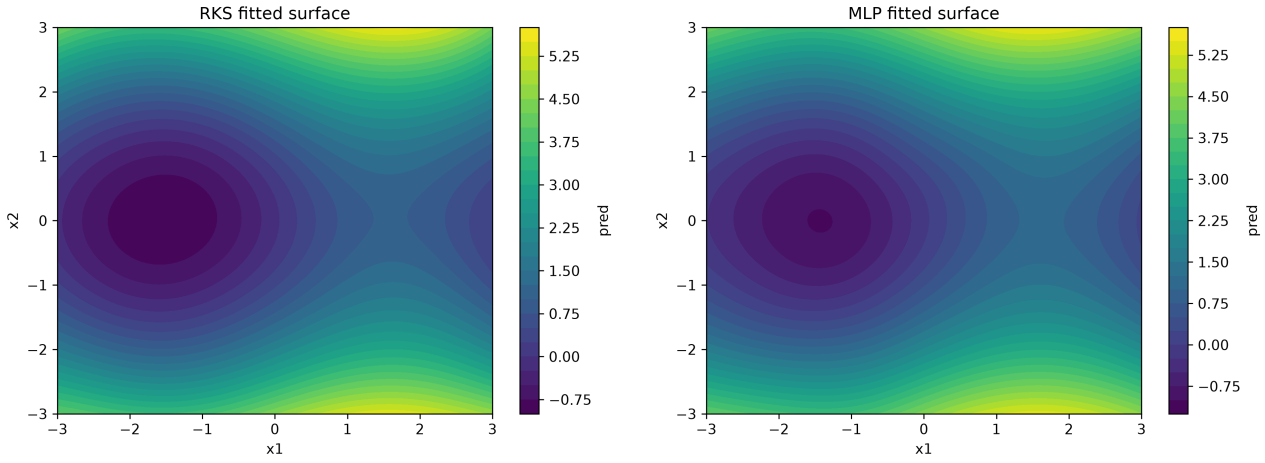


Figure 8: Selected fitted functions over the same $[-3, 3]^2$ input domain. Color bars show predicted response values.

Both fitted surfaces recover the training scatter's broad trough near $(-1.5, 0)$, the increase toward large $|x_2|$, and smooth variation along x_1. The surfaces are similar at this scale and suppress the noise in individual training observations. This visual agreement is consistent with the similar validation losses.

For the unlabeled test set, the exported files are:

- `rks_test_predictions.csv`
- `mlp_test_predictions.csv`

Each contains exactly 200 headerless rows, one scalar prediction per original test row in the original order. No test MSE is reported because the test labels are unavailable.